CampusCycle

Smart Campus Bicycle Sharing and Mobility Management System

Project Proposal

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1. Introduction

CampusCycle is a smart bicycle-sharing platform designed specifically for university campuses. The system allows students to locate, reserve, unlock, use, and return shared bicycles through a web application. Administrators can manage bicycles, stations, rentals, maintenance, and usage analytics. The project extends beyond basic rental functionality by incorporating QR-based bicycle identification, geofencing, maintenance reporting, usage analytics, and demand prediction.

2. Problem Statement

Large university campuses often require students to travel considerable distances between hostels, academic blocks, libraries, cafeterias, sports facilities, and administrative buildings. Although bicycles can solve this last-mile mobility problem, a shared bicycle system requires efficient tracking and management. A conventional rental system does not adequately address uneven bicycle distribution, unauthorized use outside the campus, maintenance issues, and difficulty in predicting where bicycles will be needed.

3. Objectives

- Provide students with a simple platform to find and rent campus bicycles.
- Implement QR-based identification for individual bicycles.
- Display bicycle and docking-station availability on an interactive campus map.
- Maintain complete rental and return records.
- Use geofencing to detect bicycles leaving the permitted campus area.
- Allow students to report bicycle faults and maintenance problems.
- Provide administrators with fleet, usage, and maintenance analytics.
- Analyze historical rental data to predict bicycle demand at different stations.
- Recommend redistribution of bicycles between stations based on predicted demand.

4. Proposed System

The system will contain two major interfaces: a student application and an administrator dashboard. Students will be able to register, locate nearby bicycles, scan QR codes, reserve and start rides, return bicycles at approved stations, view ride history, and report issues. Administrators will be able to manage bicycles and stations, monitor active rentals, track bicycle availability, review maintenance reports, detect geofence violations, and analyze station-wise and time-wise usage.

5. Key Features

- **QR-Based Bicycle Identification:** Each bicycle will have a unique QR code used to identify and validate its current status.
- **Campus Geofencing:** A virtual boundary will trigger alerts when a tracked bicycle leaves the permitted campus area.
- **Interactive Campus Map:** Stations will display current availability.
- **Maintenance Management:** Students can report faults, while unsafe bicycles can be removed from availability.
- **Demand Prediction:** Historical rental data will be used to forecast station-level demand.
- **Smart Redistribution:** The system will recommend moving bicycles from surplus stations to locations expected to have high demand.

6. System Architecture

- **Frontend:** React.js with Tailwind CSS.
- **Backend:** Node.js and Express.js REST APIs.
- **Database:** MySQL.
- **Machine Learning:** Python, Pandas, and Scikit-learn.
- **Mapping:** Leaflet/OpenStreetMap.
- **Optional IoT:** ESP32, GPS module, and electronic lock for a physical prototype.

7. Expected Outcome

A working prototype will allow a student to discover an available bicycle, scan its QR code, start a rental, view the ride status, and return the bicycle at an approved station. Administrators will have a dashboard for fleet and maintenance management. The project will also demonstrate station-level demand prediction and redistribution recommendations.

8. Project Scope

The minimum viable product will include authentication, bicycle and station management, QR-based identification, rental/return workflows, campus map visualization, maintenance reporting, and an administrator dashboard. Advanced features such as real-time GPS tracking, electronic locks, demand prediction, and smart redistribution will be implemented as higher-level modules depending on available development time and hardware availability.

9. Conclusion

CampusCycle combines web development, databases, geospatial concepts, machine learning, and optional IoT hardware into a practical campus mobility solution. The project addresses a real transportation problem while remaining feasible as a two-person semester project. Its modular design allows the basic bicycle-sharing system to be developed first and advanced intelligence and hardware features to be added incrementally.